NTNU
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DIGITAL INFRASTRUCTURE AND CYBER SECURITY

TTM4135 - Applied cryptography and network security

Practical

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Date: 25.01.2026

Table of Contents

List of Figures **i**

List of Tables **i**

1 Statistical analysis **1**

2 Caesar analysis **2**

3 Substitution analysis **3**

4 Vigenere analysis **3**

5 Hill Cipher Analysis **4**

Bibliography **6**

List of Figures

1 Frequency Distribution 2

List of Tables

1 Individual characters (1-grams) 1

2 Di-grams (pair of letters or 2-gram) 1

3 3-gram 2

4 Top 3 words in the individual character analysis 2

1 Statistical analysis

A frequency analysis of all four ciphertexts was performed using a custom Python script and CyberChef. The script computed distributions of individual characters (1-grams), digrams (2-grams), and trigrams (3-grams). To validate the results, the outputs were cross-checked using the frequency analysis tools available in CyberChef. The results obtained from both approaches were consistent.

Tables 1, 2, and 3 present the character, digram, and trigram frequency distributions for ciphertext 0. These statistics form the basis for identifying the cipher types in the subsequent analyses.

Nr	Symbol	Count	Percentage	Nr	Symbol	Count	Percentage
0	A	23	2.37	15	P	50	5.14
1	B	26	2.67	16	Q	24	2.47
2	C	45	4.63	17	R	30	3.09
3	D	25	2.57	18	S	39	4.01
4	E	33	3.40	19	T	28	2.88
5	F	42	4.32	20	U	49	5.04
6	G	48	4.94	21	V	26	2.67
7	H	21	2.16	22	W	41	4.22
8	I	21	2.16	23	X	38	3.91
9	J	32	3.29	24	Y	33	3.40
10	K	39	4.01	25	Z	19	1.95
11	L	35	3.60	26	,	45	4.63
12	M	38	3.91	27	.	35	3.60
13	N	23	2.37	28	-	24	2.47
14	O	40	4.12				

Table 1: Individual characters (1-grams)

Nr	2-gram	Count	Percentage (%)
0	UG	14	1.44
1	WC	14	1.44
2	SO	11	1.13
3	YF	11	1.13
4	ML	10	1.03
5	QO	9	0.93
6	,P	9	0.93
7	OP	8	0.82
8	PK	8	0.82
9	.X	8	0.82
10	VP	8	0.82
11	PU	8	0.82
12	EU	8	0.82
13	XF	8	0.82
14	.J	7	0.72
15	U,	7	0.72

Table 2: Di-grams (pair of letters or 2-gram)

Nr	3-gram	Count	Percentage (%)
0	UGS	4	0.41
1	GSO	4	0.41
2	FOP	4	0.41
3	OUG	3	0.31
4	WCM	3	0.31
5	UG.	3	0.31
6	GJ	3	0.31
7	VPU	3	0.31
8	RKY	3	0.31
9	KYF	3	0.31
10	MLQ	3	0.31

Table 3: 3-gram

2 Caesar analysis

The 1-gram distributions of ciphertexts 0–3 were compared against the English character frequency distribution provided in the Appendix. Each ciphertext was then tested against all 29 possible Caesar shifts using the custom Python script.

By examining the character frequencies, ciphertext 2 was found to produce a distribution closely matching English under a single Caesar shift, and the decrypted output was immediately readable. In English, the most frequent character is **E**, while ciphertext 2 has **A** as its most frequent symbol (14.51%). This suggests a mapping from plaintext **E** to ciphertext **A**.

Using the 29-character alphabet defined in the assignment, this corresponds to a Caesar shift of 25 (equivalently, a shift of 4 in the opposite direction). Applying this shift using the tool *Caesar cipher: Encode and decode online* 2026 resulted in the plaintext shown below, confirming that ciphertext 2 was encrypted using a Caesar cipher.

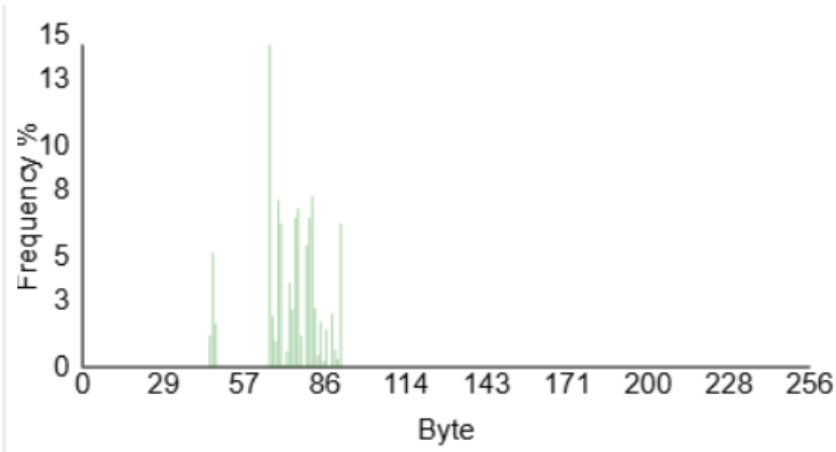

Figure 1: Frequency Distribution

Nr	Symbol	Count	Percentage (%)
0	A	164	14.51
1	B	26	2.30
2	C	13	1.15

Table 4: Top 3 words in the individual character analysis

```
EXCEPTIONALLYNICENATURE.
    IFHIRSTHADLOOKEDATHERINSTEADOFLOOKINGINTENTLYATHISTEACUPHEWOULDHAVESEENHELENBLUSH
    ,PARTLYWITHPLEASURE,
    PARTLYWITHANIMPULSEOFAFFECTIONTOWARDSTHEYOUNGMANWHOHADSEEMED,
    ANDWOULDSEEMAGAIN,SOUGLYANDSOLIMITED.SHEPITIEDHIM,
    FORSHESUSPECTEDTHATHESUFFERED,ANDSHEWASINTERESTEDINHIM,
    FORMANYOFTHETHINGSHESAIDSEEMEDTOHERTRUESHEADMIREDTHEMORALITYOFYOUTH
    ,ANDYETSHEFELTIMPRISONED.
    ASIFHERINSTINCTWERETOESCAPETOSOMETHINGBRIGHTLYCOLOUREDANDIMPERSONAL
    ,WHICHSHECOULDHOLDINHERHANDS,
    SHEWENTINTOTHEHOUSEANDRETURNEDWITHHEREMBROIDERY.
    BUTHEWASNOTINTERESTEDINHEREMBROIDERYHEDIDNOTEVENLOOKATIT.
    ABOUTMISSVINRACE,HEBEGAN,--OH,LOOKHERE,DOLETSBEST.JOHNANDHELEN
    ,ANDRACHELANDTERENCE--WHATSSHELIKEDOESSHEREASON,DOESSHEFEEL,
    ORISSHEMERELYAKINOFFOOTSTOOL.OHNO,SAIDHELEN,WITHGREATDECISION.
    FROMHEROBSERVATIONSATTEASHEWASINCLINEDTODOUBTWHETHERHIRSTWASTHEPERSONTOEDUCATERACHE
    .SHEHADGRADUALLYCOMETOBEINTERESTEDINHERNIECE,
    ANDFONDOFHERSHEDISLIKEDSOMETHINGABOUTHERVERYMUCH,
    SHEWASAMUSEDBYOTHERSBUTSHEFELTHER,ONTHEWHOLE,
    ALIVEIFUNFORMEDHUMANBEING,EXPERIMENTAL,
    ANDNOTALWAYSFORTUNATEINHEREXPERIMENTS,BUTWITHPOWERSOFSOMEKIND,
    ANDACAPACITYFORFEELING.SOMEWHEREINTHE
```

3 Substitution analysis

Ciphertext 1 was identified as a random simple substitution cipher based on its statistical properties. Its 1-gram frequency distribution resembles English, but no single Caesar shift produces a consistent alignment for all characters, ruling out a Caesar cipher. Additionally, the digram and trigram distributions exhibit clear English-like structure, which is characteristic of a simple substitution cipher.

According to the Appendix, the most frequent English digram is `TH`, followed by `HE`, and the most frequent trigram is `THE`. In ciphertext 1, the most frequent digram is `LP`, the second most frequent digram is `PK`, and the most frequent trigram is `LPK`.

Although the most frequent trigram is not necessarily `THE`, the mapping `LP` $\rightarrow$ `TH` and `PK` $\rightarrow$ `HE` is consistent across 1-gram, 2-gram, and 3-gram statistics. In particular, the trigram `LPK` has the same overlapping structure as `THE`. Furthermore, the letter `K` has a high frequency (13.65%), which is consistent with the high frequency of `E` in English.

Based on this combined evidence, ciphertext 1 is classified as a random simple substitution cipher, with `LPK` corresponding to `THE`.

4 Vigenere analysis

The Index of Coincidence (IC) was computed for all four ciphertexts using *Index of Coincidence Calculator* 2026. Ciphertexts 1 and 2 exhibited IC values close to 0.065, consistent with monoalphabetic ciphers. Ciphertexts 0 and 3 had lower IC values, indicating polyalphabetic encryption.

Given the known period of 5, ciphertext$\tilde{3}$ was split into five interleaved streams. The IC for each stream ranged between 0.060 and 0.068, which is characteristic of English plaintext, confirming that ciphertext 3 was encrypted using a Vigenère cipher.

Each stream was then treated as a Caesar cipher and analyzed using frequency matching. The resulting shifts were 2, 6, 11, 17, and 13, corresponding to the key letters **CGLRN** (using A=0 indexing). Decrypting ciphertext 3 with this key produced coherent English plaintext, confirming the result.

Decoded using *Vigenère cipher: Encode and decode online* 2026:

```
GREEnTREESTOODoVErTHEMASIFITWEREAMoVINGFORCEHELDATREST.
    ASSHEAPpRoACHED,SHECOuLDHEArEVELYNSVOICEREpEATINGMONOToNOUSLy,
    HERETHEn--HERE--GoODDOGGIE,
    COMEHEREFORAMOMENTNOTHINGSEEMEDTOHApPENITALLSTOODSTILL,
    ANDTHENSHEREALISEDTHATONEoFTHEFIGURESwASHELENAMBrOSEANDTHEDUSTAGAINBEGANTOSETTLE
    .THEGROupInDEEDHADCOMETOGETHERINAMISCELlANEOUSWAYOnETEA-
    TABlEJOININGTOANOTHERTEA-TABlE,ANDDECK-
    CHAIRSSErVINGTOCONNECTTWOGROUPS.
    BUTEVENATADISTANCEITCOULDBESEENTHATMrS.FLUSHING,
    UPRIGHTANDIMPERIOuS,DOMINATEDTHEPARTy.
    SHEWASTALKINGVEHEMENTlYTOHELENACROSSTHETABLE.
    TEnDAYSUNDERCANVAS,SHEWASSAYING.nOCOMFORTS.IFYOUWANTCOMFORTS,
    DoNTCOME.BUTIMAYTELLYoU,IFYOuDOnTCOMEyOUlLREGRETITALLYOURLIFE.
    yOUSAyYESATTHISMOMEnTMrS.FLUSHINGCAUGHTSIGHToFRACHEL.AH,
    THERESYOURNIECE.SHESPrOMISED.yOURECOMInG,
    ARENTYOUHAVInGADOpTEDTHEPLAn,SHEpURSUEDITWITHTHEENERGyOFACHILD
    .RACHELTOOKHErPARTWITHEAGERNESS.OFCoURSEIMCOMING.SoAREYoU,
    HELEN.AnDMr.pEPpERTOO.
    ASSHESATSHEREAlISEDTHATSHEWASSURROUnDEDBYPEOPLESHEKNEW,
    BUTTHATTERENCEWASNOTAMONGTHEM.
    FROMVARIOUSANGlESPEOpLEBEGANSAYINGWHATTHEyTHOUGHTOFTHEPROPOSEDEXPEDITIoN
    .ACCORDInGTOSoMEITWOuLDBE
```

5 Hill Cipher Analysis

After identifying ciphertexts 1, 2, and 3 as random substitution, Caesar, and Vigenère ciphers , the remaining ciphertext 0 was selected as the first one out to test if it was Hill cipher.

Step 1: Candidate digrams A custom Python script was used to compute digram frequencies for ciphertext 0. The two most frequent ciphertext digrams were found to be `UG` and `WC`. According to the Appendix, the most common English digrams are `TH` and `HE`. Since a Hill cipher preserves digram structure under a linear transformation, it is reasonable to assume that frequent plaintext digrams correspond to frequent ciphertext digrams. Based on this observation, the candidate mappings `TH` $\rightarrow$ `UG` and `HE` $\rightarrow$ `WC` were selected for testing.

Step 2: Solving for the key Assuming a 2×2 Hill cipher and using the 29-character alphabet defined in the assignment (A–Z, comma, period, dash), the Hill encryption equation $C = K \cdot P \pmod{29}$ was applied to the assumed digram correspondences. The Python script converted the letters to numerical values using the mapping A=0, B=1, ..., and solved the resulting system of linear equations modulo 29. This produced the encryption matrix

$$K = \begin{pmatrix} 8 & 6 \\ 24 & 2 \end{pmatrix} \pmod{29}.$$

Written row by row and converted back to characters, this matrix corresponds to the key `IGYC`.

Step 3: Verification by decryption To verify the candidate key, the Python script was used to compute the modular inverse of K and decrypt ciphertext 0 in digram blocks. The resulting plaintext was immediately recognizable as coherent English, beginning with

```
WEREFIXEDSOSTEADILYUPONABALLONTHERAILOFTHESHIPTHATSHEWOULDHAVEBEENSTARTLEDANDANNOYE
.SHEHADBEGUNHERMEDITATIONSWITHASHOUTOFLAUGHTER,
CAUSEDBYTHEFOLLOWINGTRANSLATIONFROMTRISTANINSHRINKINGTREPIDATIONHISSHAMEHESEEMSTOHI
-LIKEBRIDE.SEEMSITSOSENSELESSWHATISAYSHECRIEDTHATITDID,
ANDTHREWDOWNTHEBOOK.NEXTSHEHADPICKEDUPCOWPERSLETTERS,
THECLASSICPRESCRIBEDBYHERFATHERWHICHHADBOREDHER,
SOTHATONESENTENCECHANCINGTOSAYSOMETHINGABOUTTHESMELLOFBROOMINHISGARDEN
,
SHEHADTHEREUPONSEENTHELITTLEHALLATRICHMONDLADENWITHFLOWERSONTHEDAYOFHERMOTHERSFUNER
,SMELLINGSOSTRONGTHATNOWANYFLOWER-
SCENTBROUGHTBACKTHESICKLYHORRIBLESENSATIONANDSOFROMONESCENESHEPASSED
,HALF-HEARING,HALF-SEEING,TOANOTHER.
SHESAWHERAUNTLUCYARRANGINGFLOWERSINTHEDRAWING-ROOM.AUNTLUCY,
SHEVOLUNTEERED,IDONTLIKETHESMELLOFBROOMITREMINDSMEOFFUNERALS.
NONSENSE,RACHEL,AUNTLUCYREPLIEDDONTSAYSUCHFOOLISHTHINGS,DEAR.
IALWAYSTHINKITAPARTICULARLYCHEERFULPLANT.
```

The successful recovery of readable English confirms that ciphertext 0 was encrypted using a 2×2 Hill cipher with key `IGYC`.

Bibliography

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